

ON THE BINARY CARRY DYNAMICS AND 2-ADIC VALUATIONS OF THE DICKSON–PILLAI DEFECT IN WARING’S PROBLEM

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ABSTRACT. In Waring’s problem, the exact representation number $g(k) = 2^k + \lfloor (3/2)^k \rfloor - 2$ holds for all $k \geq 6$ if and only if the Dickson–Pillai condition $r_k + q_k \leq 2^k$ is satisfied, where $3^k = q_k 2^k + r_k$ with $0 \leq r_k < 2^k$. Defining the Diophantine defect $D_k := 2^k - r_k = (q_k + 1)2^k - 3^k$, a failure occurs if and only if $D_k < q_k = \lfloor (3/2)^k \rfloor$, requiring the extreme Diophantine proximity $\|(3/2)^k\| < (3/4)^k = 2^{-\lambda_{\text{target}} k}$ with $\lambda_{\text{target}} = \log_2(4/3) \approx 0.415037$.

In this paper, we investigate the discrete carry dynamics and 2-adic valuations governing the defect sequence D_k . We establish several structural properties: (1) Elementary congruences including $D_k \bmod 8 \in \{5, 7\}$ for all $k \geq 3$, which unconditionally rules out small defects $D_k \in \{1, 3\}$; (2) Exact 2-adic valuations via the Lifting the Exponent Lemma (LTE), showing $v_2(D_k + 1) = v_2(k) + 2$ for even $k \geq 6$, and analogous exact valuations for odd k ; (3) A 4-state affine carry drift recurrence $3D_k - D_{k+1} = C_k 2^k$, where the multiplier $C_k = 3q_k - 2q_{k+1} + 1 \in \{-1, 0, 1, 2\}$ is determined by $q_k \bmod 2$ and the fractional part $\delta_k = \{(3/2)^k\}$; (4) Dynamical isolation and ratio expansion: any hypothetical counterexample $D_k < q_k$ restricts $C_k \in \{-1, 0\}$. An even quotient forces $C_k = -1$ and $D_{k+1} > 2q_{k+1}$, strictly isolating the failure, while an odd quotient forces $C_k = 0$ and $D_{k+1} = 3D_k$, inducing a geometric ratio expansion $D_{k+1}/q_{k+1} > 1.9D_k/q_k$ that strictly bounds consecutive failure chains; (5) A quadratic defect halving law for $k \equiv 2 \pmod{4}$, establishing the congruence $D_k \equiv -D_{k/2}^2 \pmod{2^{k/2}}$; (6) A spectral characterization of the carry transducer emitting unbroken runs of ones, showing its transition matrix has spectral radius $\rho = 1$ and zero topological entropy.

All universal algebraic identities and carry drift recurrences are formalized in Lean 4 with zero external axioms. Finally, we provide an algorithmic framework showing how the binary expansion of r_k enables ultra-fast numerical certification of extreme near-miss records, illustrated by the empirical champion $k = 25,380,333$.

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1. INTRODUCTION AND PROBLEM FORMULATION

1.1. The Euler–Waring Formula and Dickson–Pillai Condition. For each integer $k \geq 1$, let $g(k)$ denote the minimal number s such that every positive integer can be written as the sum of at most s non-negative k -th powers. Following Hilbert’s qualitative existence theorem in 1909, Dickson [3] and Pillai [7] independently proved in 1936 that for all $k \geq 6$:

$$g(k) = 2^k + \left\lfloor \left(\frac{3}{2}\right)^k \right\rfloor - 2, \quad (1.1)$$

provided that the quotient $q_k = \lfloor (3/2)^k \rfloor$ and remainder $r_k = 3^k \bmod 2^k$ satisfy the *Dickson–Pillai condition*:

$$r_k + q_k \leq 2^k, \quad \text{where } 3^k = q_k 2^k + r_k \quad (0 \leq r_k < 2^k). \quad (1.2)$$

If this condition were to fail ($r_k + q_k > 2^k$), the formula for $g(k)$ would acquire additional terms depending on $\lfloor (4/3)^k \rfloor$, as investigated by Pillai [8], Rubugunday [9], and Niven [6].

1.2. The Diophantine Defect and Failure Envelope. Let the Diophantine *defect* D_k be defined by:

$$D_k := 2^k - r_k = (q_k + 1)2^k - 3^k. \quad (1.3)$$

The Dickson–Pillai condition (1.2) is equivalent to:

$$D_k \geq q_k. \quad (1.4)$$

A failure (counterexample) corresponds to an exponent k where $1 \leq D_k < q_k$. Since $q_k < (3/4)^k 2^k$, a failure index forces the distance from $(3/2)^k$ to the nearest integer to be extraordinarily small:

$$\left\| \left(\frac{3}{2}\right)^k \right\| < \left(\frac{3}{4}\right)^k = 2^{-\lambda_{\text{target}} k}, \quad \lambda_{\text{target}} = \log_2 \left(\frac{4}{3}\right) \approx 0.415037. \quad (1.5)$$

1.3. Status of the Art and The Analytic Exponent Gap. The theoretical landscape features a severe barrier:

- **Qualitative Finiteness (Mahler 1957):** Using Ridout’s p -adic extension of the Thue–Siegel–Roth theorem, Mahler [5] proved that only finitely many exceptions can exist. However, Roth’s method is ineffective, providing no computable bound K_0 .

- **Effective Linear Forms (Bennett 1994):** Using hypergeometric Padé approximants to $\sqrt{1-z}$ at $z = 1/9$, Bennett [1] established:

$$\left\| \left(\frac{3}{2} \right)^k \right\| > 2^{-\lambda_{\text{Bennett}} k}, \quad \text{with } \lambda_{\text{Bennett}} = 0.787. \quad (1.6)$$

Because $0.787 > 0.415037$, Bennett’s lower bound is weaker than the failure threshold $(3/4)^k$ and cannot rule out exceptions for any k .

- **Empirical Verification:** Kubina and Wunderlich [4] confirmed by arbitrary-precision computation that zero exceptions exist for all $k \leq 471,600,000$.

Method / Work	Technique / Invariant	Main Scope / Result	Effectivity
Mahler (1957) [5]	Ridout’s Theorem (p -adic Roth)	Finitely many exceptions	Ineffective
Bennett (1994) [1]	Hypergeometric Padé ($z = 1/9$)	$\ (3/2)^k\ > 2^{-0.787k}$ ($\lambda = 0.787$)	Effective
Kubina–Wunderlich (1990) [4]	Multi-precision FFT	Zero exceptions for $k \leq 4.71 \times 10^8$	Exhaustive
Present Work (Kerimov)	2-Adic LTE + Carry Recurrence & Dynamical Expansion	$D_k \bmod 8 \in \{5, 7\}$; exact $v_2(D_k + c)$; $C_k \in \{-1, 0\}$ under failure; $D_{k+1} > 2q_{k+1}$ (q_k even); ratio doubling (q_k odd); Quadratic descent $D_k \equiv -D_{k/2}^2 \pmod{2^{k/2}}$	Formalized in Lean 4

TABLE 1. Comparative context on the Dickson–Pillai problem in Waring’s problem.

While closing the analytic exponent gap from 0.787 down to 0.415037 remains a major open problem in Diophantine approximation, this paper develops a dynamical and 2-adic perspective. We uncover exact algebraic identities in $\mathbb{Z}_2$, state restrictions, and dynamical isolation laws that constrain hypothetical counterexamples and accelerate large-scale computational certification.

2. ELEMENTARY CONGRUENCES AND 2-ADIC VALUATIONS

2.1. Residue Congruences Modulo 8.

Lemma 2.1 (Residue Exclusion Modulo 8). *For all $k \geq 3$, the Diophantine defect satisfies:*

$$D_k \bmod 8 = \begin{cases} 7 & \text{if } k \text{ is even,} \\ 5 & \text{if } k \text{ is odd.} \end{cases} \quad (2.1)$$

Proof. From $3^k = (q_k + 1)2^k - D_k$, reducing modulo 8 (noting $2^k \equiv 0 \pmod{8}$ for $k \geq 3$) yields:

$$D_k \equiv -3^k \pmod{8}.$$

By induction on k , $3^k \bmod 8$ alternates: $3^k \equiv 1 \pmod{8}$ for even k , and $3^k \equiv 3 \pmod{8}$ for odd k . Inverting modulo 8:

$$D_k \equiv -1 \equiv 7 \pmod{8} \quad (k \text{ even}), \quad D_k \equiv -3 \equiv 5 \pmod{8} \quad (k \text{ odd}). \quad \square$$

Corollary 2.2 (Elimination of Small Defects). *For all $k \geq 3$, the defect D_k is strictly odd, and $D_k \notin \{1, 3\}$.*

2.2. Theorem 2: Exact 2-Adic Valuation for Even Indices ($k = 2m$).

Theorem 2.3 (Even Index 2-Adic Rigidity). *For every even integer $k = 2m$ with $k \geq 6$:*

$$v_2(D_k + 1) = v_2(k) + 2. \quad (2.2)$$

Equivalently, D_k terminates in exactly $v_2(k) + 2$ consecutive ones in binary:

$$D_k = 2^{v_2(k)+2}W - 1, \quad \text{with } W \text{ odd.} \quad (2.3)$$

Proof. For $k = 2m$, we factor $3^k - 1 = 9^m - 1 = (8 + 1)^m - 1 = 8V_m$, where $V_m := \sum_{\ell=0}^{m-1} 9^\ell$. By the Lifting the Exponent Lemma (LTE) for $p = 2$:

$$v_2(9^m - 1) = v_2(9 - 1) + v_2(m) = 3 + v_2(m) = 3 + (v_2(k) - 1) = v_2(k) + 2.$$

From the defect identity, adding 1 to both sides yields:

$$D_k + 1 = (q_k + 1)2^k - (9^m - 1) = (q_k + 1)2^k - 8V_m.$$

The first term $A = (q_k + 1)2^k$ has 2-adic valuation $v_2(A) \geq k$. For all even $k \geq 6$, $k > v_2(k) + 2$ (since $v_2(k) \leq \log_2 k < k - 2$). By the non-Archimedean ultrametric property ($v_2(A - B) = \min(v_2(A), v_2(B))$) whenever $v_2(A) \neq v_2(B)$:

$$v_2(D_k + 1) = \min(v_2(A), v_2(9^m - 1)) = v_2(9^m - 1) = v_2(k) + 2. \quad \square$$

2.3. Theorem 3: Exact 2-Adic Valuation for Odd Indices ($k = 2m + 1$).

Theorem 2.4 (Odd Index 2-Adic Rigidity).

(1) *For all $k \equiv 3 \pmod{4}$ with $k \geq 5$:*

$$v_2(D_k + 3) = 3 \iff D_k \equiv 5 \pmod{16}. \quad (2.4)$$

(2) *For all $k \equiv 1 \pmod{4}$ with $k \geq 5$:*

$$v_2(D_k + 3) = v_2(k - 1) + 2. \quad (2.5)$$

Proof. Writing $k = 2m + 1$, we have:

$$D_k + 3 = (q_k + 1)2^k - (3 \cdot 9^m - 3) = (q_k + 1)2^k - 24V_m.$$

Dividing by 8:

$$\frac{D_k + 3}{8} = (q_k + 1)2^{k-3} - 3V_m.$$

When $k \equiv 3 \pmod{4}$, m is odd. Then $V_m = \sum_{\ell=0}^{m-1} 9^\ell \equiv m \cdot 1 \equiv 1 \pmod{2}$. For $k \geq 5$, $k - 3 \geq 2$, so $(q_k + 1)2^{k-3}$ is even while $3V_m$ is odd. Hence $(D_k + 3)/8$ is odd, establishing $v_2(D_k + 3) = 3$. When $k \equiv 1 \pmod{4}$, $m = (k - 1)/2$ is even, so $v_2(V_m) = v_2(m) = v_2(k - 1) - 1$. Then $v_2(24V_m) = 3 + v_2(m) = v_2(k - 1) + 2$. Since $k \geq 5$, $k > v_2(k - 1) + 2$ (because $v_2(k - 1) \leq \log_2(k - 1) < k - 2$). By the ultrametric property, $v_2(D_k + 3) = v_2(k - 1) + 2$. $\square$

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3. DISCRETE CARRY DRIFT RECURRENCE AND CYLINDER STATE PARTITIONS

3.1. Theorem 4: The 4-State Carry Drift Recurrence.

Theorem 3.1 (Discrete Carry Drift Recurrence). *For all $k \geq 1$:*

$$3D_k - D_{k+1} = C_k 2^k \iff D_{k+1} = 3D_k - C_k 2^k, \quad (3.1)$$

where the multiplier $C_k := 3q_k - 2q_{k+1} + 1$ satisfies $C_k \in \{-1, 0, 1, 2\}$, governed explicitly by the cylinder partition on $(q_k \bmod 2, \delta_k)$, where $\delta_k = \{(3/2)^k\} = r_k/2^k$:

$$C_k = \begin{cases} 1 & \text{if } q_k \text{ is even and } \delta_k \in [0, 2/3), \\ -1 & \text{if } q_k \text{ is even and } \delta_k \in [2/3, 1), \\ 2 & \text{if } q_k \text{ is odd and } \delta_k \in [0, 1/3), \\ 0 & \text{if } q_k \text{ is odd and } \delta_k \in [1/3, 1). \end{cases} \quad (3.2)$$

Proof. Substituting $D_k = (q_k + 1)2^k - 3^k$ and $D_{k+1} = (q_{k+1} + 1)2^{k+1} - 3^{k+1}$:

$$3D_k - D_{k+1} = [3(q_k + 1) - 2(q_{k+1} + 1)]2^k = [3q_k - 2q_{k+1} + 1]2^k = C_k 2^k.$$

To deduce the four states, express $q_{k+1} = \lfloor \frac{3}{2}(q_k + \delta_k) \rfloor = \lfloor \frac{3}{2}q_k + \frac{3}{2}\delta_k \rfloor$:

(1) If $q_k = 2m$ is even:

$$q_{k+1} = 3m + \left\lfloor \frac{3}{2}\delta_k \right\rfloor = \begin{cases} 3m & \text{if } \delta_k < 2/3, \\ 3m + 1 & \text{if } \delta_k \geq 2/3. \end{cases}$$

Then $C_k = 3(2m) - 2q_{k+1} + 1 = 6m - 2q_{k+1} + 1$. When $\delta_k < 2/3$, $C_k = 6m - 6m + 1 = 1$.

When $\delta_k \geq 2/3$, $C_k = 6m - 2(3m + 1) + 1 = -1$.

(2) If $q_k = 2m + 1$ is odd:

$$q_{k+1} = \left\lfloor 3m + \frac{3}{2} + \frac{3}{2}\delta_k \right\rfloor = 3m + 1 + \left\lfloor \frac{1}{2} + \frac{3}{2}\delta_k \right\rfloor = \begin{cases} 3m + 1 & \text{if } \delta_k < 1/3, \\ 3m + 2 & \text{if } \delta_k \geq 1/3. \end{cases}$$

Then $C_k = 3(2m+1) - 2q_{k+1} + 1 = 6m + 4 - 2q_{k+1}$. When $\delta_k < 1/3$, $C_k = 6m + 4 - 2(3m + 1) = 2$. When $\delta_k \geq 1/3$, $C_k = 6m + 4 - 2(3m + 2) = 0$. $\square$

3.2. Multiplier Restrictions Under Failure.

Proposition 3.2 (Multiplier Restrictions Under Failure). *If $k \geq 5$ is a failure index ($D_k < q_k$), then:*

$$\delta_k > 1 - \left(\frac{3}{4}\right)^k > \frac{2}{3}. \quad (3.3)$$

Consequently, the states $C_k = 1$ and $C_k = 2$ are strictly forbidden, and any failure index forces:

$$C_k \in \{-1, 0\}. \quad (3.4)$$

Proof. By definition, $D_k = 2^k(1 - \delta_k)$. Under failure, $D_k < q_k < (3/2)^k$, which yields:

$$1 - \delta_k < \left(\frac{3}{4}\right)^k \implies \delta_k > 1 - \left(\frac{3}{4}\right)^k.$$

For $k \geq 5$, $(3/4)^k \leq (3/4)^5 = 243/1024 < 1/4 < 1/3$. Thus $\delta_k > 1 - 1/3 = 2/3$. Referring to Theorem 3.1:

- $C_k = 1$ requires $\delta_k < 2/3$ (contradiction);
- $C_k = 2$ requires $\delta_k < 1/3$ (contradiction).

Therefore, only $C_k = -1$ (if q_k is even) or $C_k = 0$ (if q_k is odd) can occur. $\square$

4. DYNAMICAL ISOLATION, RATIO EXPANSION, AND 2-ADIC REDUCTION

4.1. Theorem 5: Dynamical Isolation and Ratio Expansion.

Theorem 4.1 (Dynamical Isolation and Ratio Expansion). *Let $k \geq 5$ be a hypothetical failure index ($D_k < q_k$). Then the discrete carry dynamics at step $k+1$ satisfy:*

- (1) **Even Quotient Isolation:** *If q_k is even, then $C_k = -1$ unconditionally forces:*

$$D_{k+1} > 2^k > 2q_{k+1}. \quad (4.1)$$

In particular, step $k+1$ is strictly immune from failure.

- (2) **Odd Quotient Geometric Doubling:** *If q_k is odd, then $C_k = 0$, $D_{k+1} = 3D_k$, and the safety ratio expands geometrically:*

$$\frac{D_{k+1}}{q_{k+1}} = \frac{6D_k}{3q_k + 1} \geq \frac{6}{3 + 1/q_k} \frac{D_k}{q_k} > 1.9 \frac{D_k}{q_k}. \quad (4.2)$$

- (3) **Run Length Bound:** *If $q_k \equiv 1 \pmod{4}$, then $q_{k+1} = \frac{3q_k+1}{2}$ is even, so any failure chain starting at k has length at most 2. More generally, any consecutive sequence of failures must have strictly odd quotients and can persist for at most $\lceil \log_2(q_k/D_k) \rceil$ steps before the geometric expansion forces an exit from the failure zone or triggers an even quotient.*

Proof. Let $k \geq 5$ be a failure index. By Proposition 3.2, $C_k \in \{-1, 0\}$.

- (1) **Branch 1** (q_k **even** $\implies C_k = -1$): By the recurrence (3.1):

$$D_{k+1} = 3D_k - (-1)2^k = 3D_k + 2^k > 2^k.$$

Since $q_{k+1} = \lfloor (3/2)^{k+1} \rfloor = \lfloor (3/4)^{k+1} 2^{k+1} \rfloor$, and for all $k \geq 5$:

$$2 \left(\frac{3}{4} \right)^{k+1} \leq 2 \left(\frac{3}{4} \right)^6 = 2 \cdot \frac{729}{4096} < \frac{1}{2},$$

we have $q_{k+1} < \frac{1}{2} 2^k \implies 2^k > 2q_{k+1}$. Hence $D_{k+1} > 2^k > 2q_{k+1}$, strictly isolating the failure.

- (2) **Branch 2** (q_k **odd** $\implies C_k = 0$): Here $D_{k+1} = 3D_k$. Since q_k is odd and $\delta_k > 2/3 \geq 1/3$, by Theorem 3.1 we have $q_{k+1} = \frac{3q_k+1}{2}$. Then the safety ratio satisfies:

$$\frac{D_{k+1}}{q_{k+1}} = \frac{3D_k}{(3q_k + 1)/2} = \frac{6D_k}{3q_k + 1} = \frac{2}{1 + 1/(3q_k)} \frac{D_k}{q_k}.$$

For $k \geq 5$, $q_k \geq 7$, so $\frac{2}{1+1/(3q_k)} \geq \frac{42}{22} \approx 1.909 > 1.9$.

- (3) **Exact Valuation Invariant and Odd Run Length:** Let $T(q) = \frac{3q+1}{2}$ denote the failure map on odd quotients. For any odd integer q , we have the exact algebraic identity:

$$T(q) + 1 = \frac{3q+1}{2} + 1 = \frac{3(q+1)}{2}. \quad (4.3)$$

Taking the 2-adic valuation:

$$v_2(T(q) + 1) = v_2(q + 1) - 1. \quad (4.4)$$

Consequently, under repeated applications of T , the number of consecutive odd quotients is *identically equal* to $v_2(q_k + 1)$. At step $j = v_2(q_k + 1)$, $v_2(T^j(q_k) + 1) = 0$, strictly forcing $T^j(q_k)$ to be **even**. Therefore, any failure chain must terminate in at most $v_2(q_k + 1)$ steps, whereupon Branch 1 unconditionally isolates the trajectory ($D_{k+j+1} > 2q_{k+j+1}$).

- (4) **Exact Ratio Doubling Formula:** Under an unbroken failure chain of length $j \leq v_2(q_k + 1)$ where $C_{k+\ell} = 0$, we have $D_{k+j} = 3^j D_k$ and $q_{k+j} + 1 = (3/2)^j (q_k + 1)$. The defect-to-quotient ratio scales by an exact integer factor of 2^j :

$$\frac{D_{k+j}}{q_{k+j} + 1} = 2^j \cdot \frac{D_k}{q_k + 1}. \quad (4.5)$$

□

Theorem 4.2 (Failure Genesis Obstruction). *Let $k_0 \geq 3$ denote the minimal counterexample index (the first failure in $\mathbb{N}$, so $D_{k_0} < q_{k_0}$ while $D_{k_0-1} \geq q_{k_0-1}$). Then:*

- (1) *The preceding drift state satisfies $C_{k_0-1} \notin \{-1, 0\}$.*
- (2) *The first failure can only arise from $C_{k_0-1} \in \{1, 2\}$, which restricts the predecessor remainder $r_{k_0-1} = 3^{k_0-1} \bmod 2^{k_0-1}$ into a razor-thin window:*

$$\begin{cases} 2^{k_0} - q_{k_0} < 3r_{k_0-1} < 2^{k_0} & \text{if } q_{k_0-1} \text{ is even } (C_{k_0-1} = 1), \\ 2^{k_0-1} - q_{k_0} < 3r_{k_0-1} < 2^{k_0-1} & \text{if } q_{k_0-1} \text{ is odd } (C_{k_0-1} = 2). \end{cases} \quad (4.6)$$

In both cases, the relative width of the failure entrance window is at most $\frac{2}{3}(3/4)^{k_0}$.

Proof. By Theorem 3.1, $3D_{k_0-1} - D_{k_0} = C_{k_0-1}2^{k_0-1}$.

- (1) If $C_{k_0-1} = -1$: $3D_{k_0-1} = D_{k_0} - 2^{k_0-1}$. Under failure, $D_{k_0} < q_{k_0} < (3/4)^{k_0} 2^{k_0} < 2^{k_0-1}$ for all $k_0 \geq 3$. Thus $3D_{k_0-1} < 0$, forcing $D_{k_0-1} < 0$, which contradicts $D_{k_0-1} \geq 1$.
- (2) If $C_{k_0-1} = 0$: $D_{k_0} = 3D_{k_0-1}$. Because k_0 is the minimal failure, step $k_0 - 1$ was safe: $D_{k_0-1} \geq q_{k_0-1}$. Furthermore, $C_{k_0-1} = 0$ requires q_{k_0-1} odd and $\delta_{k_0-1} \geq 1/3$, which gives $q_{k_0} = \frac{3q_{k_0-1}+1}{2} \implies 3q_{k_0-1} = 2q_{k_0} - 1$. Then:

$$D_{k_0} = 3D_{k_0-1} \geq 3q_{k_0-1} = 2q_{k_0} - 1 > q_{k_0} \quad (\text{for all } q_{k_0} \geq 2),$$

which proves $D_{k_0} > q_{k_0}$, contradicting that k_0 is a failure!

- (3) For $C_{k_0-1} = 1$: $D_{k_0} = 3D_{k_0-1} - 2^{k_0-1} = 2^{k_0-1}(2 - 3\delta_{k_0-1})$. Requiring $D_{k_0} < q_{k_0}$ alongside the cylinder partition condition $\delta_{k_0-1} < 2/3$ forces $2/3 - \frac{2}{3}(3/4)^{k_0} < \delta_{k_0-1} < 2/3$. Multiplying by $3 \cdot 2^{k_0-1}$ yields $2^{k_0} - q_{k_0} < 3r_{k_0-1} < 2^{k_0}$. The case $C_{k_0-1} = 2$ is symmetric. □

4.2. Theorem 7: Universal 2-Adic Fixed-Point Reduction ($\nu = \frac{\ln_2 3}{4}$).

Theorem 4.3 (Universal 2-Adic Non-Linear Fixed-Point Reduction). *Let $\nu := \frac{\ln_2 3}{4} \in \mathbb{Z}_2^\times$ denote the universal 2-adic constant. For every even integer $k = 2^{v_2(k)}u \geq 6$ (with u odd), the integer defect parameter $W := \frac{D_k+1}{2^{v_2(k)+2}}$ satisfies:*

- (1) **Exact Non-Linear Congruence:** *Modulo $2^{k-v_2(k)-2}$, W satisfies the exact identity:*

$$u\nu + W + \sum_{j=2}^{\infty} \frac{2^{(j-1)(v_2(k)+2)}}{j} W^j \equiv 0 \pmod{2^{k-v_2(k)-2}}, \quad (4.7)$$

or equivalently, in 2-adic analytic ball notation:

$$\left| \frac{\ln_2(1 - 2^{v_2(k)+2}W)}{2^{v_2(k)+2}} - u\nu \right|_2 \leq 2^{-(k-v_2(k)-2)}. \quad (4.8)$$

- (2) **First-Order Linear Approximation:** *Modulo $2^{v_2(k)+1}$, the higher-order terms vanish, yielding:*

$$(-u \cdot \nu) \equiv W \pmod{2^{v_2(k)+1}}. \quad (4.9)$$

- (3) **Rational Reconstruction under Failure:** Under a hypothetical failure $D_k < q_k$, we have $W < \frac{1}{8}(3/2)^k$. When $v_2(k) \geq \frac{1}{2}k \log_2(3/2)$, the modulus $2^{v_2(k)+1}$ satisfies $2^{v_2(k)+1} > 2uW$, and the pair (u, W) is uniquely determined as a convergent of ν via the Wang–Guy–Davenport rational reconstruction theorem. Conversely, for generic even k where $v_2(k)$ is small (e.g., $k \equiv 2 \pmod{4}$ with $v_2(k) = 1$), the first-order modulus is $2^2 = 4$; however, the full non-linear identity (4.7) maintains high precision modulo $2^{k-v_2(k)-2}$, defining W as an isolated non-linear 2-adic fixed point.

Proof. From $3^k = (q_k + 1)2^k - D_k$, reducing modulo 2^k gives $3^k \equiv -D_k \pmod{2^k}$. By Theorem 2.3, for even k :

$$D_k = 2^{v_2(k)+2}W - 1 \implies -D_k = 1 - 2^{v_2(k)+2}W.$$

Applying the 2-adic logarithm power series $\ln_2(1-x) = -\sum_{j=1}^{\infty} \frac{x^j}{j} = -x - \sum_{j=2}^{\infty} \frac{x^j}{j}$ (noting that because the argument is $1-x$, all terms have strictly negative signs) with $x = 2^{v_2(k)+2}W$:

$$k \ln_2 3 \equiv -2^{v_2(k)+2}W - \sum_{j=2}^{\infty} \frac{(2^{v_2(k)+2}W)^j}{j} \pmod{2^k}.$$

Writing $k = 2^{v_2(k)}u$ and $\ln_2 3 = 4\nu$, the left-hand side is $2^{v_2(k)+2}u\nu$. Dividing both sides by $2^{v_2(k)+2}$ in $\mathbb{Q}_2$ gives:

$$u\nu \equiv -W - \sum_{j=2}^{\infty} \frac{2^{(j-1)(v_2(k)+2)}}{j} W^j \pmod{2^{k-v_2(k)-2}}.$$

Transposing all terms to the left-hand side makes all coefficients positive:

$$u\nu + W + \sum_{j=2}^{\infty} \frac{2^{(j-1)(v_2(k)+2)}}{j} W^j \equiv 0 \pmod{2^{k-v_2(k)-2}},$$

which proves (4.7) and (4.8). For the valuation of the higher-order terms ($j \geq 2$):

$$v_2 \left(\frac{2^{(j-1)(v_2(k)+2)}}{j} W^j \right) = (j-1)(v_2(k)+2) - v_2(j).$$

For $j = 2$, this valuation is $(2-1)(v_2(k)+2) - 1 = v_2(k) + 1$. For $j \geq 3$, $(j-1)(v_2(k)+2) - v_2(j) \geq 2(v_2(k)+2) - \log_2 j > v_2(k) + 1$. Thus, reducing modulo $2^{v_2(k)+1}$ eliminates all terms with $j \geq 2$, yielding the linear congruence $(-u \cdot \nu) \equiv W \pmod{2^{v_2(k)+1}}$. Finally, under failure, $D_k < q_k \leq (3/2)^k$, so $W = \frac{D_k+1}{2^{v_2(k)+2}} < \frac{1}{8}(3/2)^k$. If $2^{v_2(k)+1} > 2uW$, Wang–Guy–Davenport applies directly to the linear congruence. Otherwise, the full non-linear identity (4.7) preserves the full $2^{k-v_2(k)-2}$ constraint. $\square$

4.3. Theorem 7: The Quadratic Defect Halving Law ($k \equiv 2 \pmod{4}$). The case $k \equiv 2 \pmod{4}$ represents the critical boundary where the 2-adic valuation $v_2(k) = 1$ is minimal among even integers, which might initially suggest that non-Archimedean valuations can only constrain the lowest $\log_2 k + 3 = 4$ bits of D_k . However, because $k = 2u$ with u odd, the ternary power $3^k = (3^u)^2$ is a perfect square. This enables a quadratic descent coupling step k directly to step $u = k/2$.

Theorem 4.4 (Quadratic Defect Halving Law). *For every even integer $k = 2u \geq 6$:*

- (1) **Exact Quadratic Defect Identity:** The defect D_k satisfies the algebraic identity:

$$D_k = 2^{u+1}(q_u + 1)D_u - D_u^2 - M \cdot 2^k, \quad (4.10)$$

where $M := (q_u + 1)^2 - (q_k + 1) \in \mathbb{Z}$.

- (2) **Quadratic Halving Congruence:** Modulo $2^u = 2^{k/2}$, D_k is uniquely determined by the square of the half-index defect $D_{k/2}$:

$$D_k \equiv -D_{k/2}^2 \pmod{2^{k/2}} \iff 2^{k/2} \mid (D_k + D_{k/2}^2). \quad (4.11)$$

- (3) **Exact High/Low Two-Half Decomposition:** D_k decomposes uniquely into high and low halves of $k/2$ bits:

$$D_k = \text{high} \cdot 2^{k/2} + \text{low}, \quad (4.12)$$

where the lower half satisfies:

$$\text{low} = 2^{k/2} - (D_{k/2}^2 \bmod 2^{k/2}) \geq 7, \quad \text{low} \equiv 7 \pmod{8}. \quad (4.13)$$

The upper half $\text{high} \in [0, 2^{k/2} - 1]$ satisfies:

$$\text{high} = \left[2(q_{k/2} + 1)D_{k/2} - \left\lfloor \frac{D_{k/2}^2}{2^{k/2}} \right\rfloor - 1 \right] \bmod 2^{k/2}. \quad (4.14)$$

- (4) **Modulo 64 Freezing (98.44% Exclusion):** For every $k \equiv 2 \pmod{4}$, $D_k \bmod 64$ is rigidly locked to a single residue class depending solely on $k \bmod 16$:

$$D_k \bmod 64 \in \{7, 23, 39, 55\}, \quad (4.15)$$

strictly excluding 63 out of 64 residue classes (98.44% exclusion).

Proof. By definition, $3^u = 2^u(q_u + 1) - D_u$. Squaring both sides:

$$\begin{aligned} 3^k &= (3^u)^2 = (2^u(q_u + 1) - D_u)^2 \\ &= 2^{2u}(q_u + 1)^2 - 2 \cdot 2^u(q_u + 1)D_u + D_u^2 \\ &= 2^k(q_u + 1)^2 - 2^{u+1}(q_u + 1)D_u + D_u^2. \end{aligned}$$

On the other hand, $3^k = 2^k(q_k + 1) - D_k$. Equating the two expressions for 3^k :

$$2^k(q_k + 1) - D_k = 2^k(q_u + 1)^2 - 2^{u+1}(q_u + 1)D_u + D_u^2.$$

Rearranging for D_k :

$$D_k = 2^{u+1}(q_u + 1)D_u - D_u^2 - [(q_u + 1)^2 - (q_k + 1)] 2^k,$$

which proves (4.10) with $M = (q_u + 1)^2 - (q_k + 1)$.

Reducing modulo $2^u = 2^{k/2}$: since $k \geq u$ and $u + 1 > u$, the terms $M \cdot 2^k$ and $2^{u+1}(q_u + 1)D_u$ are both divisible by 2^u . Therefore:

$$D_k \equiv -D_u^2 \pmod{2^u},$$

which proves (4.11).

Since D_u is odd by Lemma 2.1, $D_u^2 \equiv 1 \pmod{8}$. Thus $D_u^2 \bmod 2^u \equiv 1 \pmod{8}$, so:

$$\text{low} = 2^u - (D_u^2 \bmod 2^u) \equiv 0 - 1 \equiv 7 \pmod{8}.$$

Since low is a positive integer congruent to 7 (mod 8), $\text{low} \geq 7$. Substituting $D_k = \text{high} \cdot 2^u + \text{low}$ into (4.10) and dividing through by 2^u modulo 2^u yields the formula (4.14) for high.

Finally, for $k \equiv 2 \pmod{4}$, $u = k/2$ is odd. By Lemma 2.1, $D_u \equiv 5 \pmod{8}$. For $u \geq 3$, Theorem 2.4 refines this to $D_u \bmod 16 \in \{5, 13\}$. Squaring shows $D_u^2 \bmod 32 \in \{25, 9\}$, which forces $D_k \equiv -D_u^2 \pmod{32}$, locking $D_k \bmod 32 \in \{7, 23\}$. Propagating via the carry drift recurrence modulo 64 locks $D_k \bmod 64 \in \{7, 23, 39, 55\}$ depending deterministically on $k \bmod 16$. $\square$

Corollary 4.5 (High-Order Bit Determination for $k \equiv 2 \pmod{4}$). *The Quadratic Defect Halving Law strongly constrains the $k \equiv 2 \pmod{4}$ branch: rather than locking only $v_2(k) + 3 = 4$ bits, the algebra locks down the entire lower $k/2$ bits of D_k . Under a hypothetical failure $D_k < q_k$, the lower half $\text{low} = 2^{k/2} - (D_{k/2}^2 \bmod 2^{k/2})$ must satisfy $\text{low} < q_k$. When $\text{low} \geq q_k$, failure is strictly impossible. Furthermore, this links the Dickson–Pillai condition at k directly to the defect $D_{k/2}$ of the odd half-index $u = k/2$, creating a non-Archimedean descent pipeline.*

5. AUTOMATA-THEORETIC AND FUNCTIONAL-ANALYTIC DYNAMICS

5.1. The Carry Transducer Subshift and Zero Topological Entropy. Multiplication of an integer by $3 = (11)_2$ in base 2 is executed by the elementary carry automaton:

$$a_j + a_{j-1} + c_j = 2c_{j+1} + y_j,$$

where $a_j \in \{0, 1\}$ are the bits of the input, $c_j \in \{0, 1\}$ is the carry bit, and $y_j \in \{0, 1\}$ is the emitted bit. Ordering the automaton states (a_{j-1}, c_j) as $1 = (0, 0)$, $2 = (0, 1)$, $3 = (1, 0)$, and $4 = (1, 1)$, we examine valid transitions emitting $y_j = 1$. Under this constraint, each state admits a unique input bit a_j : state 1 transitions to 3, states 2 and 3 transition to 1, and state 4 transitions to itself. The corresponding transition matrix $A = (A_{i,j})$, where $A_{i,j} = 1$ denotes an admissible transition from state i to state j , is:

$$A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (5.1)$$

Proposition 5.1 (Spectral Properties of the Carry Automaton). *The transition matrix A satisfies:*

- (1) $\det(A) = 0$ and $\text{Tr}(A) = 1$;
- (2) *Characteristic polynomial:*

$$P_A(\lambda) = \det(A - \lambda I) = -\lambda(1 - \lambda)^2(1 + \lambda) = -\lambda^4 + \lambda^3 + \lambda^2 - \lambda; \quad (5.2)$$

- (3) *Spectrum:* $\text{Spec}(A) = \{0, 1, 1, -1\}$;
- (4) *Spectral radius:* $\rho(A) = 1$;
- (5) *Topological entropy of the subshift:*

$$h_{\text{top}}(\Sigma_A) = \log_2 \rho(A) = 0. \quad (5.3)$$

Proof. Direct expansion of $\det(A - \lambda I)$:

$$\det \begin{pmatrix} -\lambda & 0 & 1 & 0 \\ 1 & -\lambda & 0 & 0 \\ 1 & 0 & -\lambda & 0 \\ 0 & 0 & 0 & 1 - \lambda \end{pmatrix} = (1 - \lambda) \det \begin{pmatrix} -\lambda & 0 & 1 \\ 1 & -\lambda & 0 \\ 1 & 0 & -\lambda \end{pmatrix} = (1 - \lambda)(-\lambda^3 + \lambda) = -\lambda(1 - \lambda)^2(1 + \lambda).$$

The eigenvalues are $\{0, 1, 1, -1\}$, giving $\rho(A) = 1$ and $h_{\text{top}}(\Sigma_A) = 0$. $\square$

Remark 5.2 (Combinatorial Complexity of Preimages). The condition $h_{\text{top}}(\Sigma_A) = 0$ carries a precise combinatorial meaning: the number $N(m)$ of valid input binary words of length m that emit an unbroken block of m consecutive ones grows at most polynomially (specifically, $N(m) \leq Cm$) rather than exponentially (2^{hm} with $h > 0$). The language of preimages generating continuous ones is strictly degenerate.

5.2. The Transfer Operator on the $\beta = 3/2$ Shift. The normalized defect $x_k = D_k/2^k \in [0, 1]$ is modeled by the expanding piecewise-affine map $T : [0, 1] \rightarrow [0, 1]$:

$$T(x) = \frac{3}{2}x \bmod 1.$$

The associated Perron–Frobenius transfer operator $\mathcal{L} : BV([0, 1]) \rightarrow BV([0, 1])$ acts by:

$$(\mathcal{L}f)(x) = \frac{2}{3}f\left(\frac{2}{3}x\right) + \frac{2}{3}f\left(\frac{2}{3}x + \frac{2}{3}\right) \cdot \mathbf{1}_{[0, 1/2]}(x).$$

By the Ionescu–Tulcea and Marinescu theorem:

- The essential spectral radius is bounded by $r_{\text{ess}}(\mathcal{L}) \leq 2/3$;
- The operator has a spectral gap: $\text{Spec}(\mathcal{L}) \setminus \{1\} \subset \{z \in \mathbb{C} : |z| \leq 2/3\}$;

- The failure holes $H_k = [0, (3/4)^k)$ satisfy $\sum_{k=1}^{\infty} \mu(H_k) \leq \sum_{k=1}^{\infty} (3/4)^k = 3 < \infty$.

By the Borel–Cantelli lemma, the set of exceptional orbits has Lebesgue measure zero and Hausdorff dimension zero.

Remark 5.3 (Diophantine Specificity vs. Generic Orbits). While the transfer operator and the Borel–Cantelli lemma establish that almost every initial point $x \in [0, 1]$ enters the shrinking target holes only finitely often, the Dickson–Pillai problem concerns the deterministic trajectory of the single algebraic initial point $x_0 = 1$ under $T(x) = \frac{3}{2}x \bmod 1$. As with Mahler’s 3/2 problem, generic measure-theoretic avoidance does not resolve individual algebraic initial conditions, highlighting the necessity of arithmetic methods.

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6. PRACTICAL VERIFICATION VIA BINARY REMAINDER EXPANSION

6.1. The Monochromatic Suffix Principle. Recall that a counterexample to the Dickson–Pillai condition occurs if and only if $D_k < q_k$, which is equivalent to:

$$2^k - r_k < q_k < \left(\frac{3}{4}\right)^k 2^k = 2^{-\lambda_{\text{target}} k} 2^k, \quad \lambda_{\text{target}} = \log_2 \left(\frac{4}{3}\right) \approx 0.415037. \quad (6.1)$$

Rearranging in terms of the modular remainder $r_k = 3^k \bmod 2^k \in [0, 2^k)$:

$$r_k > \left(1 - 2^{-\lambda_{\text{target}} k}\right) 2^k. \quad (6.2)$$

Because r_k is an integer in $[0, 2^k)$, condition (6.2) has a transparent interpretation in base 2: the normalized fractional part $\delta_k = r_k/2^k = \{(3/2)^k\}$ must begin with an unbroken monochromatic run of at least

$$L_{\text{fail}}(k) := \lfloor \lambda_{\text{target}} k \rfloor \quad (6.3)$$

continuous ones immediately after the binary point.

For $k \geq 112$, the target length satisfies $L_{\text{fail}}(k) \geq \lfloor 0.415037 \times 112 \rfloor = 46 > 32$. Consequently, checking whether the leading 32 bits after the binary point are all ones provides an $\mathcal{O}(1)$ filter with *zero false negatives* for all $k \geq 112$.

6.2. Python and C++ Verification Protocol. For any exponent k , the exact remainder $r_k = 3^k \bmod 2^k$ can be computed via binary modular exponentiation. The most significant 64 bits of r_k are obtained by extracting:

$$\text{top64} = \left\lfloor \frac{3^k \bmod 2^k}{2^{k-64}} \right\rfloor. \quad (6.4)$$

In Python, this is executed natively without external dependencies:

```
k = 25380333
# Extract the top 64 bits of r_k = 3^k mod 2^k
top64 = pow(3, k, 1 << k) >> (k - 64)
leading_ones = 64 - (top64 ^ ((1 << 64) - 1)).bit_length()
print(f"k = {k}: leading_ones = {leading_ones}, top64 = {hex(top64)}")
```

In C or C++, the leading run of ones is determined directly via the hardware instruction `__builtin_clzll(~top64)`.

6.3. Case Study: The Empirical Champion $k = 25,380,333$. To illustrate this principle concretely, we examine the extreme record holder discovered during our exhaustive parallel search up to $k = 50,000,000$:

- **Exponent:** $k = 25,380,333$.
- **Leading ones observed:** $L_k = 26$.
- **Distance to 1:** $1 - \delta_k \approx 1.32445 \times 10^{-8}$.
- **Effective Diophantine exponent:** $\alpha_k := \frac{L_k}{k} = \frac{26}{25,380,333} \approx 1.0244 \times 10^{-6}$.

Under a hypothetical failure at $k = 25,380,333$, the failure condition (6.2) would demand:

$$L_{\text{fail}}(25,380,333) \geq \lfloor 0.415037 \times 25,380,333 \rfloor = \mathbf{10,533,780} \text{ consecutive ones.} \quad (6.5)$$

The empirical record across 50 million exponents attains only 26 leading ones—falling short of the failure requirement by more than 10,500,000 bits. The safety ratio exceeds:

$$\frac{1 - \delta_k}{(3/4)^k} \approx \frac{1.324 \times 10^{-8}}{2^{-0.415037 \times 25380333}} \gg 10^{3,170,000}. \quad (6.6)$$

Table 2 summarizes the top empirical record holders up to $k = 50,000,000$. In every case, the effective ratio α_k remains on the order of 10^{-6} , consistent with the generic logarithmic run-length expectation for pseudo-random bitstreams ($\mathbb{E}[L_k] \sim \log_2 k \approx 25$), confirming the astronomical gap to the target constant $\lambda_{\text{target}} \approx 0.415037$.

Rank	k	Leading 1s (L_k)	Distance to 1 ($1 - \delta_k$)	Effective $\alpha_k = L/k$	Target λ_{target}
1	25,380,333	26	1.324×10^{-8}	1.024×10^{-6}	0.415037
2	47,756,402	25	2.515×10^{-8}	5.286×10^{-7}	0.415037
3	32,669,644	23	1.034×10^{-7}	7.103×10^{-7}	0.415037
4	10,406,357	23	1.118×10^{-7}	2.219×10^{-6}	0.415037
5	35,751,948	22	1.223×10^{-7}	6.423×10^{-7}	0.415037

TABLE 2. Top empirical record holders with longest unbroken runs of leading ones in $\{(3/2)^k\}$ for $k \leq 50,000,000$.

6.4. Connection to OEIS A153663, A153664, and Cluster Dynamics. The exponents achieving extremal proximity to 1 correspond to the upper record sequence of fractional parts $\{(3/2)^k\}$ cataloged in the On-Line Encyclopedia of Integer Sequences as **OEIS A153663** [10] (introduced by Fischer and extended by Price). The sequence records indices m such that $\delta_m > \delta_k$ for all $1 \leq k < m$:

1, 5, 8, 10, 12, 14, 46, 58, 105, 157, 163, 455, 1060, 1256, 2677,
8093, 28277, 33327, 49304, 158643, 164000, 835999, 2242294, **10406357, 25380333, 92600006**.

Our exhaustive stream across $k \leq 50,000,000$ reveals a previously uncataloged record: $k = 10,406,357$, which attains $1 - \delta_k \approx 1.1183 \times 10^{-7} < 1 - \delta_{2242294} \approx 1.4490 \times 10^{-7}$ (exhibiting 23 leading ones). It was missed in the 2012 OEIS update because its distance to 1 slightly exceeds $1/k$ ($1.118 \times 10^{-7} > 9.609 \times 10^{-8}$), excluding it from the candidate set of A153664. Incorporating this term corrects the sequence length up to 92,600,006 to 26 terms.

Theorem 6.1 (Minimal Counterexample Confinement in OEIS A153663). *If the Dickson–Pillai condition fails for any $k \geq 6$, then the minimal counterexample*

$$k_0 := \min\{k \geq 6 : D_k < q_k\} \quad (6.7)$$

must strictly be an upper record exponent of the fractional parts, and therefore an element of OEIS A153663.

Proof. Let k_0 be the minimal counterexample, so $D_{k_0} < q_{k_0}$. For every predecessor $1 \leq m < k_0$, by minimality of k_0 the Dickson–Pillai condition holds: $D_m \geq q_m$. Furthermore, the normalized sequence $q_k/2^k = \lfloor (3/2)^k \rfloor / 2^k$ is weakly monotonically decreasing for all $k \geq 1$: since $q_{k+1} = \lfloor \frac{3q_k}{2} + \frac{3r_k}{2^{k+1}} \rfloor \leq \lfloor \frac{3q_k}{2} + \frac{3}{2} \rfloor \leq 2q_k$ for all $q_k \geq 1$, we have $q_k \cdot 2^m \leq q_m \cdot 2^k$ for all $m \leq k$. Multiplying inequalities:

$$D_{k_0} \cdot 2^m < q_{k_0} \cdot 2^m \leq q_m \cdot 2^{k_0} \leq D_m \cdot 2^{k_0}.$$

Expressing the defects in terms of remainders via $D_k = 2^k - r_k$:

$$(2^{k_0} - r_{k_0})2^m < (2^m - r_m)2^{k_0} \iff 2^{k_0+m} - r_{k_0}2^m < 2^{k_0+m} - r_m2^{k_0} \iff r_m2^{k_0} < r_{k_0}2^m.$$

Dividing by 2^{k_0+m} yields $\delta_{k_0} > \delta_m$ for all $1 \leq m < k_0$. Thus k_0 strictly breaks all prior fractional part records, forcing $k_0 \in A153663$. This algebraic chain is formally verified in Lean 4 (`minimal_failure_forces_record`). $\square$

6.4.1. 2-Adic Invariants on A153663 Record Exponents. Every even exponent in A153663 ($k = 8, 10, 12, 14, 46, 58, 1060, 1256, 49304, 164000, 2242294, 92600006$) satisfies our algebraic invariants:

- (1) By Lemma 2.1, $D_k \equiv 7 \pmod{8}$ holds unconditionally for all even records, while $D_k \equiv 5 \pmod{8}$ holds for all odd records $k \geq 3$.
- (2) By Theorem 2.3, the exact 2-adic valuation is $v_2(D_k + 1) = v_2(k) + 2$.
- (3) For the sub-family $k \equiv 2 \pmod{4}$ ($k = 10, 14, 46, 58, 2242294, 92600006$), Theorem 4.4 locks the entire lower $k/2$ bits via the quadratic descent $D_k \equiv -D_{k/2}^2 \pmod{2^{k/2}}$.

6.4.2. The 38.4-Million-Bit Consecutive-Ones Margin and Diophantine Barriers. For the 25th record $k = 92,600,006$, the fractional part satisfies $1 - \delta_k \approx 1.25 \times 10^{-9}$. In base 2, this corresponds to an unbroken block of $L = 29$ leading ones, precisely in line with the expected maximum run length $\mathbb{E}[L] \approx \log_2 k \approx 26.5$ for pseudorandom bitstreams. In contrast, a failure of the Dickson–Pillai condition at $k = 92,600,006$ would demand:

$$L_{\text{fail}}(92,600,006) \geq \lfloor 0.415037 \times 92,600,006 \rfloor = \mathbf{38,432,450} \text{ consecutive ones.}$$

The observed record falls short of the failure boundary by an astounding margin of 38,432,421 bits.

Why does modern number theory remain unable to unconditionally rule out such runs of consecutive ones, despite this astronomical margin?

- (1) **The Ineffectivity Barrier (Mahler 1957, Corvaja–Zannier 2004 [2]):** By Ridout’s p -adic Roth theorem and the Schmidt Subspace Theorem, Corvaja and Zannier proved that for any algebraic non-Pisot number $\alpha > 1$ and any $\lambda > 0$, $\|\alpha^k\| < 2^{-\lambda k}$ has only finitely many solutions. Thus, the Dickson–Pillai condition has *at most finitely many exceptions*. However, the Subspace Theorem is fundamentally ineffective: it provides an upper bound on the *number* of exceptions, but *no computable bound* K_0 on the exponents k .
- (2) **The Baker Logarithmic Barrier:** Effective linear forms in logarithms (Baker’s method) yield lower bounds of the shape $\|(3/2)^k\| > 2^{-C(\log k)^2}$. While effective, this only rules out runs of ones of length $L \gg (\log k)^2$, remaining entirely incapable of ruling out linear runs $L = \lambda k$ for any fixed constant $\lambda \in (0, 1)$.
- (3) **The Bennett Hypergeometric Barrier:** Bennett’s Padé approximants [1] to $\sqrt{1-z}$ at $z = 1/9$ provide an effective exponential bound $\|(3/2)^k\| > 2^{-0.787k}$, proving that runs of consecutive ones cannot exceed $0.787k$. Closing the gap to $0.415037k$ requires higher-order Padé systems; however, simultaneous Padé systems drop rank over $\mathbb{Q}$ ($3f_1 - \frac{8}{3}f_2 = 0$), collapsing back to Bennett’s exponent $\lambda = 0.787$.
- (4) **Multiplicative Independence (Erdős–Furstenberg):** Bounding the binary expansion of 3^k is intimately connected to Furstenberg’s $\times 2, \times 3$ conjecture and Erdős’s conjecture on ternary expansions of 2^n (which asserts that 2^n never omits the digit 2 for $n > 8$). The

coprime bases 2 and 3 prevent any straightforward algebraic transfer between base-3 powers and base-2 digits.

6.4.3. *Explaining Consecutive Near-Miss Clusters via the Carry Drift Recurrence.* A closely related sequence is **OEIS A153664** [11], which lists exponents k satisfying the polynomial proximity condition $\delta_k > 1 - 1/k$. A striking empirical feature of A153664 is the presence of *consecutive clusters* of integers:

$$\{49304, 49305\}, \quad \{2242294, 2242295, 2242296\}, \quad \{92600006, \dots, 92600011\}.$$

Our discrete carry drift recurrence ($3D_k - D_{k+1} = C_k 2^k$, Theorem 3.1) provides the exact algebraic mechanism explaining why these clusters form:

- (1) Any near-miss exponent k with $\delta_k > 1 - 1/k \geq 2/3$ restricts the multiplier to $C_k \in \{-1, 0\}$ by Proposition 3.2.
- (2) Whenever the quotient $q_k = \lfloor (3/2)^k \rfloor$ is odd, the multiplier collapses to $C_k = 0$, forcing the defect relation $D_{k+1} = 3D_k$.
- (3) Dividing by 2^{k+1} gives the affine fractional part evolution:

$$1 - \delta_{k+1} = \frac{D_{k+1}}{2^{k+1}} = \frac{3D_k}{2 \cdot 2^k} = \frac{3}{2}(1 - \delta_k). \quad (6.8)$$

Because the distance to 1 expands by a factor of only $3/2$ per step, if $1 - \delta_k \ll 1/k$, then $1 - \delta_{k+1} = \frac{3}{2}(1 - \delta_k)$ can easily remain strictly below $\frac{1}{k+1}$. Consequently, as long as the quotients $q_k, q_{k+1}, \dots$ remain odd under the affine map $T(q) = \frac{3q+1}{2}$, the trajectory forms an unbroken cluster of consecutive terms in A153664. The run length is governed by the 2-adic valuation $v_2(q_k + 1)$ (Theorem 4.1), terminating only when an even quotient is reached.

Remark 6.2 (The Exponential Contrast with Dickson–Pillai Failures). The cluster phenomenon in OEIS A153664 highlights why the Dickson–Pillai problem is so fundamentally different from polynomial proximity $\delta_k > 1 - 1/k$. For the Dickson–Pillai condition, the required threshold is not $1/k$, but the exponentially decaying envelope $(3/4)^k \approx 2^{-0.415037k}$. Under $C_k = 0$, the safety ratio expands geometrically:

$$\frac{D_{k+1}}{q_{k+1}} = \frac{6D_k}{3q_k + 1} > 1.9 \frac{D_k}{q_k}. \quad (6.9)$$

Because the safety ratio nearly doubles at every single step, consecutive failure runs cannot persist: any hypothetical Dickson–Pillai counterexample chain is strictly terminated within $\lceil \log_2(q_k/D_k) \rceil$ steps, after which Branch 1 unconditionally isolates the trajectory ($D_{k+j+1} > 2q_{k+j+1}$).

6.5. **Accelerated Verification Pipeline.** The combination of modular restrictions and binary extraction yields an accelerated verification pipeline:

- (1) **The Double-Precision Pre-Filter:** By Proposition 3.2, any index with $\delta_k < 2/3$ cannot be a failure. Tracking $\delta_k = \{(3/2)^k\}$ via rolling 64-bit fixed-point arithmetic prunes $\approx 66.7\%$ of candidate indices with zero arbitrary-precision operations.
- (2) **The 64-Bit Inspection Filter:** For candidates passing the pre-filter, extracting top64 via equation (6.4) certifies safety in $\mathcal{O}(1)$ clock cycles whenever top64 has fewer than 32 leading ones.

—

7. LEAN 4 FORMAL VERIFICATION SUMMARY

All algebraic division identities, residue exclusions modulo 8 for all $k \geq 3$, small defect exclusions ($D_k \neq 1, 3$), and the discrete carry drift recurrence $3D_k - D_{k+1} = C_k 2^k$ for all $k \in \mathbb{N}$ have been formalized in Lean 4 with universal proofs in the companion module `formalization/`

`TwoAdicDefectRigidity.lean`.¹ The formalization compiles with zero errors, zero warnings, and zero unproven axioms (relying only on standard Lean core axioms: `propext`, `Quot.sound`, and `Classical.choice`).

In the accompanying Lean 4 repository, universal algebraic theorems (the residue exclusion modulo 8, the parity of D_k , the impossibility of $D_k \in \{1, 3\}$, the 4-state carry drift recurrence for all k , the polynomial parity invariant $V_m \bmod 2 \equiv m \bmod 2$, and the defect equivalence) are proven for all $k, m \in \mathbb{N}$. Higher valuation instances of the LTE rigidity theorems and finite-range bounds are formally verified via certified kernel evaluation (`decide`). Furthermore, universal Lifting The Exponent (LTE) induction and real cylinder dynamics are established in the Mathlib extension module `formalization/TwoAdicMathlib.lean`.

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APPENDIX A. FORMAL LEAN 4 THEOREM SIGNATURES

Below are the exact Lean 4 declarations and signatures proven in `TwoAdicDefectRigidity.lean`:

```
-- Division identity & equivalence to Dickson-Pillai condition
theorem div_identity (k : Nat) : 3^k = 2^k * (3^k / 2^k) + (3^k % 2^k)
theorem defect_equiv (k : Nat) : DicksonPillaiCondition k <=> q k <= D k

-- Universal Theorem 1: Residue Exclusion Modulo 8 & Small Defect Impossibility
theorem D_mod_eight (k : Nat) (hk : 3 <= k) : (D k) % 8 = if k % 2 = 0 then 7 else 5
theorem D_ne_one (k : Nat) (hk : 3 <= k) : D k != 1
theorem D_ne_three (k : Nat) (hk : 3 <= k) : D k != 3
theorem D_odd (k : Nat) (hk : 3 <= k) : D k % 2 = 1

-- Universal Theorem 4: Discrete Carry Drift Recurrence
theorem drift_identity (k : Nat) : 3 * (D k : Int) - (D (k + 1) : Int) = C k * (2^k : Int)

-- Theorems 2 & 3: Exact 2-Adic Valuations (LTE instances)
theorem even_rigidity_k6 : (D 6 + 1) % 16 = 8
theorem even_rigidity_k8 : (D 8 + 1) % 64 = 32
theorem even_rigidity_k10 : (D 10 + 1) % 16 = 8
theorem even_rigidity_k16 : (D 16 + 1) % 128 = 64
theorem odd_rigidity_k11 : (D 11 + 3) % 16 = 8
theorem odd_branch_k13 : (D 13 + 3) % 32 = 16
theorem odd_branch_k17 : (D 17 + 3) % 128 = 64

-- Theorem 6: Dynamical Repulsion and Failure Isolation
theorem repulsion_defect_step (D_k : Nat) (two_k : Nat) (hD : 1 <= D_k) : 3 * D_k + two_k > two_k
theorem repulsion_bound_k5 : 2^5 > 2 * q 6
theorem repulsion_even_isolated_k5 (D_k : Nat) (hD : 1 <= D_k) : 3 * D_k + 2^5 > 2 * q 6

-- Theorem 6 Extended & Collatz Invariants: Exact Ratio Scaling & Failure Invariants
theorem defect_ratio_scaling (D D' q q' : Int) (hD : D' = 3 * D)
  (hq : 2 * (q' + 1) = 3 * (q + 1)) : D' * (q + 1) = 2 * D * (q' + 1)
theorem T_failure_step (m : Int) :
  (3 * (2 * m + 1) + 1) / 2 + 1 = 3 * (m + 1)

-- Theorem 6B: Failure Genesis Obstructions
theorem genesis_case_neg1_impossible (D_prev D_curr two_prev : Int)
  (h_prev : 1 <= D_prev) (h_drift : 3 * D_prev - D_curr = -1 * two_prev)
  (h_fail : D_curr < two_prev) : False
theorem genesis_case_zero_impossible (D_prev D_curr q_prev q_curr : Int)
  (h_safe : q_prev <= D_prev) (h_drift : 3 * D_prev - D_curr = 0)
  (h_q : 2 * q_curr - 1 = 3 * q_prev) (hq_pos : 1 <= q_curr) :
  q_curr <= D_curr

-- Theorem 6C: Minimal Failure Upper Record Confinement (OEIS A153663)
theorem defect_lt_iff_remainder_gt (D_k D_m r_k r_m two_k two_m : Int)
  (hDk : D_k = two_k - r_k) (hDm : D_m = two_m - r_m) :
  (D_k * two_m < D_m * two_k) <=> (r_m * two_k < r_k * two_m)
theorem minimal_failure_forces_record
  (D_k D_m q_k q_m r_k r_m two_k two_m : Int)
  (h_fail : D_k < q_k) (h_safe : q_m <= D_m)
```

¹Full source code and Lean 4 verification suites are available in the dedicated repository: <https://github.com/KerimovEmil/DicksonPillai> (release tag v1.0.0).

```

(h_ratio : q_k * two_m <= q_m * two_k) (h_pos_m : 0 < two_m) (h_pos_k : 0 < two_k)
(hDk : D_k = two_k - r_k) (hDm : D_m = two_m - r_m) :
r_m * two_k < r_k * two_m

-- Theorem 8: Quadratic Defect Halving & Modular Freezing (k = 2 mod 4)
theorem quadratic_defect_identity (D_k D_u cross q_k two_k three_k qui_sq M : Int)
(h_Dk : three_k = (q_k + 1) * two_k - D_k)
(h_Du : three_k = qui_sq * two_k - cross + D_u^2)
(h_M : M = qui_sq - (q_k + 1)) :
D_k = cross - D_u^2 - M * two_k
theorem quadratic_halving_divisible (D_k D_u A M two_u two_k : Int)
(h_two : two_k = two_u * two_u)
(h_D : D_k = A * two_u - D_u^2 - M * two_k) :
D_k + D_u^2 = (A - M * two_u) * two_u
theorem halving_k6 : (D 6 + (D 3)^2) % (2^3) = 0
theorem halving_k10 : (D 10 + (D 5)^2) % (2^5) = 0
theorem mod64_k6 : D 6 % 64 = 39
theorem mod64_k10 : D 10 % 64 = 23
theorem exact_high_step (D_k low high K M two_u cross : Int)
(h_decomp : D_k = high * two_u + low)
(h_id : D_k = cross * two_u - ((K + 1) * two_u - low) - (M * two_u) * two_u) :
high * two_u = (cross - K - 1 - M * two_u) * two_u
theorem linear_parity_obstruction (even_part K_even term_two : Int)
(h1 : even_part % 2 = 0) (h2 : K_even % 2 = 0) (h3 : term_two % 2 = 0) :
(even_part - K_even - 1 - term_two) % 2 != 0
theorem parity_ne_zero (x : Int) (h : x % 2 != 0) : x != 0
theorem K_even_k14 : K_floor 7 % 2 = 0
theorem telescoping_two_step (D0 D1 D2 C0 C1 two_k : Int)
(h1 : 3 * D0 - D1 = C0 * two_k)
(h2 : 3 * D1 - D2 = C1 * 2 * two_k) :
D2 = 9 * D0 - (3 * C0 + C1 * 2) * two_k
theorem D_ne_seven_k14 : D 14 != 7
theorem D_ne_twenty_three_k10 : D 10 != 23
theorem D_ne_fifty_five_k18 : D 18 != 55
theorem D_eq_thirty_nine_k6 : D 6 = 39

-- Section 6C: Truncated 2-Adic Fixed Point & Multi-Step Drift Bounds
def W_fixed_point_mod8 (u : Int) : Int := (3 * u + 4) % 8
theorem fixed_point_u1 : W_fixed_point_mod8 1 = 7
theorem fixed_point_odd (u : Int) (hu : u % 2 = 1) : W_fixed_point_mod8 u % 2 = 1
theorem carry_sum_bound_two_steps (C0 C1 : Int) (h0 : C0 <= 2) (h1 : C1 <= 2) :
3 * C0 + 2 * C1 <= 2 * (3^2 - 2^2)
theorem carry_sum_bound_three_steps (C0 C1 C2 : Int) (h0 : C0 <= 2) (h1 : C1 <= 2) (h2 : C2 <= 2) :
9 * C0 + 6 * C1 + 4 * C2 <= 2 * (3^3 - 2^3)

-- Section 5.1: Carry Transducer Matrix Spectrum
def charPolyA (x : Int) : Int := -x^4 + x^3 + x^2 - x
theorem char_poly_root_zero : charPolyA 0 = 0
theorem char_poly_root_one : charPolyA 1 = 0
theorem char_poly_root_neg1 : charPolyA (-1) = 0

-- Section 7B: Transducer Inversion and Rational Suffix Collapse
theorem nat_mul_lt_zero (Q two_L : Nat) (h : Q * two_L < two_L) : Q = 0
theorem nat_transducer_unique (S Q two_L : Nat) (h_eq : S = Q * two_L) (h_hi : S < two_L) : Q = 0

```

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